University of Pennsylvania
ESE 2180 (Electronic, Photonic and Electromechanical Devices), 2026F

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Lab 2: Semiconductor Absorption and Emission

In this lab you will explore the optical absorption and emission properties from conventional semiconductor crystals. You can do part A and B of this lab in whatever order.

Part A: Optical Absorption

Context

Optical absorption in semiconductors is one of the main experimental tools for measuring fundamental aspects of energy band structure. It also provides useful engineering information which contributes to semiconductor selection criteria for specific device applications. This part of the lab will provide some of the flavor of both aspects.

Absorbance or Abs is a measure of the amount of light that is selectively absorbed by atoms/molecules within a substance (e.g. in a process leading to optical generation of electron-hole pairs if $E > E_g$). If there is some probability that a photon interacts with the crystal lattice, it is intuitive that a photon is more likely to interact if it passes through a larger sample with more atoms. It is apparent, then, that absorption (A or $\%A$) in the sample is dependent on the distance that the light travels through the sample. It also dependent on the wavelength of the incident light, as various crystals will absorb different wavelengths to different extents. The resulting peaks from the various absorptions allow characterization and/or quantitative analysis of the sample.

Consider a stream of photons with a specific energy ("monochromatic") is directed at a plane-parallel slab of material. Since photons can be neither created nor destroyed in this experiment, a sort of energy-dependent conservation law applies:

$$R + T + A + S = 1$$

where the capital letters are the fractions of the incident photon flux (photons per unit area per unit time) which are respectively

- (R) Reflected from the front surface
- (T) Transmitted through to some point inside the material
- (A) Absorbed by the material up to that point

- (S) Scattered from the surface by scratches, dust, imperfectly flat polishing job, etc. etc. S can be assumed zero in this lab because we are using very flat surfaces, and the roughness on them are smaller than the wavelengths we are using.

The expression for a slab of finite thickness is more complicated because reflection (and scattering) from the second surface must be taken into account – see below. The extensive variable A is the absorbance – the thicker the sample, the greater the value of A. From A we can derive the intrinsic property α, the absorption coefficient (glass has a small but nonzero α for $E =$ "red", while α is almost zero for green light). α has units of inverse length.

Since there's no such thing as an "α meter", we are again forced to approach the determination of the fundamental, microscopic, intrinsic parameter through the measurement of an extrinsic quantity. Thus, if we could measure absorbance using an "absorbance meter" – the connection between A and α involves only the thickness of the sample.

Unfortunately, the only thing that's easy to measure is the "transmittance" T, the fraction of light transmitted at a given wavelength. Assume we are looking at a slab of material with a perfectly smooth, non-reflective surface ($R = S = 0$). Then $T = 1 - A$.

The intensity through an absorbing slab follows an exponential dependence of intensity on thickness.

where $I =$ intensity and $d =$ thickness of the slab. Rearranged, this becomes:

$$-\ln(T) = -\ln(1 - A) = \alpha d$$

By measuring the transmission spectrum of a material, α can be calculated as a function of the incident photon energy (E). In indirect band gap materials, α is proportional to E, while for direct band gap materials α varies with E. By determining the functional form of $\alpha(E)$, a material can be classified as having a direct or indirect band gap. At energies greater than E_g, absorption occurs. Surface effects and internal reflections give a similar result for transmission.

Experimental Determination of Band Gap by Optical Absorption

Adapted from Jaques I. Pankove's book, *Optical Processes in Semiconductors* (Dover Publications; 2nd Revised ed. edition (November 18, 2010)).

Consider a photon incident on a sample. For the photon to be absorbed, there must be a filled state in the valence band (containing an electron to be promoted) and an empty state in the conduction band (to contain the promoted electron), separated by the photon energy.

The probability that the photon is absorbed is determined by the probability that this condition is satisfied. Quantum mechanics can be used to develop a theoretical expression for the functional form of the absorption coefficient with respect to the energy of incoming radiation, based on the density of states in both the valence and conduction bands at a given energy.

For a direct band gap material with band gap energy E_g, the absorption coefficient (α) as a function of photon energy ($h\nu$) can be described as follows:

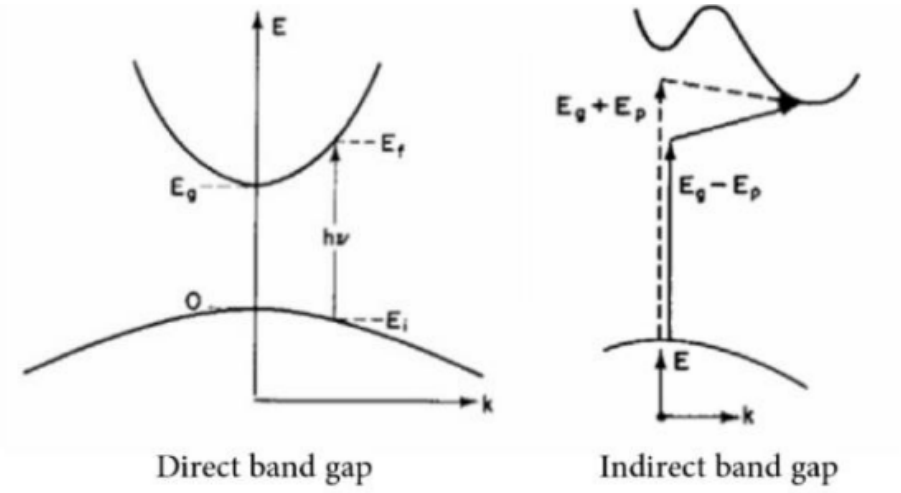
$$\alpha = A^*(h\nu - E_g)^{1/2}$$

For the indirect case, a similar expression can be used:

$$\alpha = A^*(h\nu - E_g)^2$$

A^* is a material-dependent term that is determined by the band structure of the material.

The schematic below shows the absorption process for direct band gap materials where every initial state E_i is associated with a final state E_f.



Direct band gap Indirect band gap

Materials

These are the materials that you will investigate in part A of the lab. Note the trends in dielectric constants (square of the index of refraction) with increasing cation and anion size/mass.

Material	**Refractive Index,** n	**Reflection fraction,** R	**Thickness (m)**
GaSb	3.96	0.356	7×10^{-4}
Ge	4.01	0.361	4.75×10^{-4}
Si	3.45	0.303	5×10^{-4}
GaAs	3.62	0.322	6×10^{-4}
GaP	3.33	0.290	2.1×10^{-4}
CdS	2.4	0.170	2.8×10^{-4}

Equipment

We'll use a state-of-the art dual-beam spectrophotometer (Varian / Agilent Cary 5000) to measure optical absorption in the above materials. The spectral range of the instrument is 315 nm to 3500

nm wavelength (UV to mid-infrared) corresponding to 2800 to 32000 wavenumbers (cm^{-1}), or photon energies from about 0.35 eV to 4 eV.

It is a so-called double-beam instrument. Broadband light from either of two lamps (quartz-halogen for visible and IR light, deuterium arc lamp for UV) is dispersed into a rainbow by one of two diffraction gratings and then split into two physically separated beams by a rotating disk with a hole and a mirror. So each beam is "chopped" into pulses, and the two beams are exactly 180° out of phase.

One beam goes through the sample, the other goes through nothing, and the two are recombined onto one of two detectors where the time structure of the signal allows us to record the ratio of intensities with and without the sample. This normalizes out the wavelength dependence of the light power, grating efficiency, detector efficiency, reflectivity of all the mirrors, etc.

Without this double-beam arrangement we would have to measure twice, once with the sample in and once with it out of the beam. The reason for two of everything (lamp, grating, detector) is to maintain reasonable signal levels over a broad wavelength range.

You may observe small discontinuities in your measured spectra, at wavelengths where an automatic change of lamp, grating or detector takes place. You should be aware of the following terms.

- A **baseline** is determined by measuring absorbance with no sample in the sample beam. It accounts for the wavelength dependence of all the optical components. The dual beam sample scan is then corrected for the baseline before being displayed.
- The **detector(s)** in a spectrophotometer are typically a photomultiplier tube for the UV-Vis region and a lead sulfide detector for the near infrared (NIR) region. These devices generate a voltage that is proportional to the amount of light reaching them. This voltage is then converted to an electronic signal which is relayed to the PC.
- The **lamp or source** is the generator of UV, Vis or NIR radiation (light) that passes through the monochromator before reaching the sample. Typically, a Deuterium lamp is used in the UV region and a Quartz Halogen lamp is used in the Visible/NIR region.
- The **monochromator** disperses the light from a light source or lamp into discrete wavelengths. Low-tech is a prism; high-tech is a diffraction grating.
- **Wavenumber** is the inverse of the wavelength,

$$\tilde{\nu} = \frac{1}{\lambda}.$$

  Note that this means wavenumber is proportional to energy of the radiation. Traditionally wavelength is expressed in nm but wavenumber is given in inverse centimeters, or cm^{-1}.

Procedures

We will use optical absorption methods to determine the band gaps of several semiconductor materials. We will explore column IV elements, III-V and II-VI compounds to identify and understand chemical trends in band gaps.

We want to be able to tell whether α is proportional to E^2 (indirect band gap) or $E^{1/2}$ (direct band gap), so we'll need data over a large enough range of α to make the distinction accurately.

To enhance the range of the α determination within the constraint of limited accuracy in $\%T$, $\%T(E)$ is typically measured in two or more samples of the same material with different thicknesses. This will not be necessary, however, thanks to our spectrometer's excellent linearity and dynamic range (that is, the ability to accurately measure both highly transmitting and highly absorbing samples).

Key to this performance is having a system with low photometric noise and automatically adjusting slit width (light throughput) or detector gain (light sensitivity). As α increases, the spectrometer will adjust to maintain sufficient signal, to below 0.000001% transmitted light.

The experiment consists of measuring transmission spectra for a representative sampling of semiconductor materials. You will be provided with several semiconductor samples of known thickness, mounted for easy placement in the spectrophotometer (see Materials, above).

The quantity measured (dependent variable) is $\%T$, the fraction of light transmitted at a given wavelength. As discussed previously, for bulk solid samples this depends on both absorption as the beam passes through the material and reflection of the incident beam from the front surface.

For the quantitative determination of $\alpha(E)$, calculate R from the index of refraction and assume that it doesn't vary with wavelength. Then the $\%T$ can be corrected as described above. For this experiment the sensible choice of independent variable is wavenumber (proportional to energy) rather than wavelength.

Scan parameters are provided in the table below.

Scan Range (cm^{-1})	**Step Time (s)**	**Step Size (cm^{-1})**
4000–20000	0.066	5

Part A Data processing, discussion, and analysis

Answer the following questions with short responses, max 1–2 sentences. Show calculations where requested.

(a) Why can we not assume that all light not transmitted has been absorbed? Explain why the absorption coefficient, α, is more useful than transmittance, T, when looking at different samples. **(0.5 pts)**

Transmittance alone isn't a good metric because some of the light is reflected or scattered. Thus, the absorption coefficient is more useful than transmittance because it is a definite material property, while transmittance depends on the sample thickness and other factors.

(b) Near the band edge, use the relationships below:

$$\boxed{\text{Direct: } \alpha = A^*(h\nu - E_g)^{1/2}}$$

$$\boxed{\text{Indirect: } \alpha = A^*(h\nu - E_g)^2}$$

i) What quantity should be plotted as a function of photon energy $h\nu$ to obtain a linear region? **(0.5 pts)**

Direct: α^2

Indirect: $\alpha^{1/2}$

ii) Why does an indirect optical transition require a phonon, while a direct transition does not? **(0.5 pts)**

An indirect optical transition needs a phonon to satisfy conservation of crystal momentum since the conduction band minimum is at a different k value than the valence band maximum. A direct transition does not need this since the minimum and maximum are at the same value of k.

(c) For each semiconductor in the table below, use the linearized absorption plot to estimate the bandgap from the x-intercept. You should include as much data as you collected but attempt to get to GaAs and CdS. Calculate your percent error.

Include one linearized plot showing the fitted region and intercept. In one or two sentences, explain the differences in the near-edge optical absorption in direct-gap and indirect-gap semiconductors. **(3 pts)**

Material	**Type***	**Experimental E_g (eV)**	**Known E_g (eV)**	**% Error**
GaSb	D	0.682	0.726	6.0
Ge	I	0.641	0.670	4.4
Si	I	1.125	1.120	0.4
GaAs	D	1.393	1.424	2.2
GaP	I	2.208	2.240	1.4
CdS	D	2.377	2.420	1.8

Table 1: Experimental and known bandgap energies for the semiconductor materials.

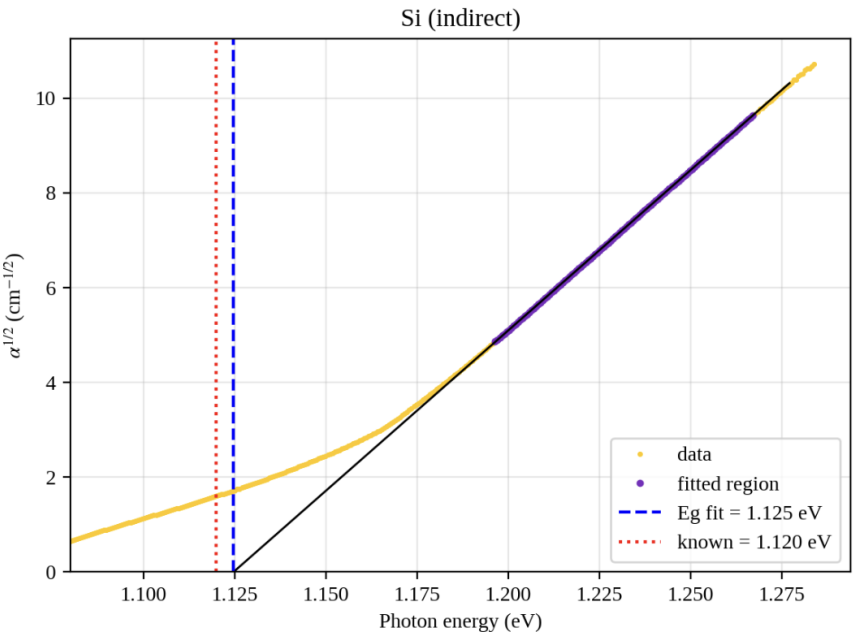


Figure 1: Sample Bandgap Estimation for Silicon Based on Data

Direct-gap materials (GaAs, CdS, GaSb) show an abrupt absorption edge, with T dropping to the noise floor within a few tens of meV, because the transition needs no phonon. Indirect-gap materials (Si, Ge, GaP) show a gradual edge spread over 0.1–0.2 eV, because the phonon-assisted process is much weaker and less instantaneous.

**D is direct bandgap, I is indirect bandgap*

$$\boxed{\text{Error} = \frac{|E_{\text{experimental}} - E_{\text{known}}|}{E_{\text{known}}} \times 100}$$

Part B: Photoluminescence

Part B Context

Photoluminescence or Emission Spectroscopy is complementary to optical absorption in semiconductors and is also an important experimental tool for measuring fundamental aspects of energy band structure.

Photoluminescence (also called fluorescence or just PL in abbreviation) gives detailed insights into recombination energies, their spread and also provides insights into excited states or defect states that contribute to emission or reduce emission.

It also provides useful engineering information which contributes to semiconductor selection criteria for specific device applications and also helps accurately measure the band gap in addition to determination of band gap type (direct vs indirect). This experiment will provide some insight into how emission spectra from semiconductors look like.

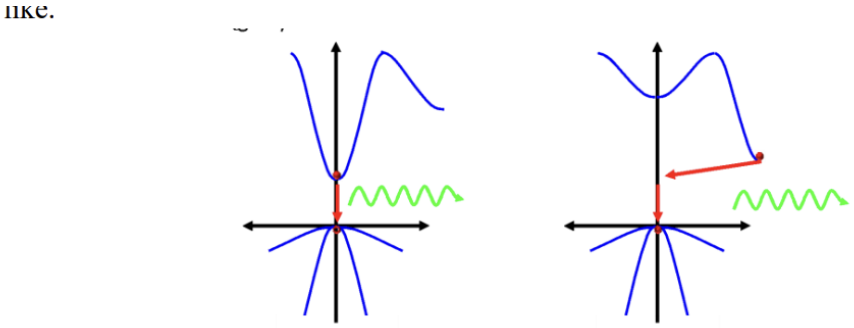


Figure 2: Direct vs Indirect gap recombination

The band structure of a material has profound effects on the applications of semiconductors in devices, particularly for optoelectronics.

- **Indirect band gap:** Need a photon and a phonon (lattice vibration/heat) to make a transition between bands. The need for both a photon and phonon simultaneously reduces the probability of occurrence.
- **Direct band gap:** Need only a photon to make a transition between bands. Good for light emitting devices (e.g., LEDs, lasers etc).

Part B Procedures

The experiment consists of measuring photoluminescence spectra for a representative sampling of semiconductor materials. You will be provided with several semiconductor samples (listed below).

Due to time constraints, you may be doing only two out of the four samples, but we will provide the complete data set. Attempt to measure as much as possible, but try to get to at least GaAs and CdS.

- GaAs (Gallium Arsenide)
- InP (Indium Phosphide)
- CdS (Cadmium Sulfide)
- ZnO (Zinc Oxide)

The semiconductor samples are mounted for easy placement in the microscope. The quantity measured is light emitted as a function of given wavelength (emission spectrum).

As discussed previously, for bulk solid samples this depends on both absorption as the beam of incident light passes through the material and its band gap. There are two types of incident light (also called excitation light): violet (405 nm) and red (633 nm).

1. Load the sample under the microscope.
2. Warm up the both lasers and open up the Labspec64 programme.
3. On the Labspec64 programme, set the spectroscopy parameters:
    - **Grating: 100.** This determines how fine your data points are.
    - **Laser type: 405 nm_Edge**
    - **Laser power: 1% or lower** (a high intensity may damage the sample)
    - **Acquisition time: 5 seconds.** This is how long the laser is turned on for and how long the spectrometer records the data. The weaker the emission of the sample, the longer you want the acquisition time to be.
    - **Accumulation: 4.** This means the programme takes 4 measurements and outputs one data set that is the normalized average of those 4.
    - **Spectro:** Which wavelength to focus on. This is 870 nm for GaAs and 510 nm for CdS. Before setting this, make sure the unit is nm.
    - **Range:** Range of wavelength to examine. Make this 450–950 nm. Make sure the box next to "Range" is selected.
    - **Title:** Give an appropriate title to your dataset – e.g. GaAs 405.
4. Focus the laser spot on a clean part of the sample.
5. Turn on the small deskside lamp and turn off the room light.
6. Run the spectrometer and record the data.
7. Switch the laser to 633 nm_Edge, repeat step 1–6, and then move on to the next sample.

Throughout the experiment and data processing later on, keep these fine points in mind.

- The two lasers, 633 nm and 405 nm, have different power outputs to begin with. The filter allows a percentage of this power output to be incident on the sample. We know that the 633 nm laser is more powerful than the 405 nm laser, so expect signals acquired with the former to be higher than signals acquired with the latter in general. This has nothing to do with the sample's interaction with the laser.

- The lasers should be focused on the sample for consistency. This should be adjusted manually every time because different samples have different thicknesses. Usually, we use the 5x lens and find the edge of sample, and change the focus until the edge is clear.
- The lasers themselves contribute to the PL data, because the spectrometer is simply detecting any intensity of light observed across a range of wavelengths. For instance, any PL measurement with the 405 nm laser will produce a peak at 405 nm that has nothing to do with the sample. This peak may or may not be large relative to the peaks produced by actual photoluminescence.
- The room light should be turned off because it floods the spectrometer with light of every wavelength in the visible spectrum. The lamp next to the spectrometer is weak enough to be turned on without compromising the data.
- Every sample has some degree of defect that may introduce several intermediate levels that sit between the valence band and the conduction band. These traps result in emission at or across wavelengths that do not correspond to the bandgap of the material. This is particularly noticeable in our CdS sample. Remember, genuine PL peaks are sharp by definition.

Part B Data processing, discussion, and analysis

You will be given eight text files containing wavelength vs intensity data from this experiment (four materials, each measured with two different lasers). Using a software of your choice (Excel, spreadsheet, MATLAB, etc.), process the data sets to produce, for each material and for each laser.

(a) Use

$$\boxed{E(\text{eV}) = \frac{1240 \text{ nm} \cdot \text{eV}}{\lambda \text{ (nm)}}}$$

to calculate the energy of each excitation laser.

$$E(405 \text{ nm}) = 3.06 \text{ eV}$$

$$E(633 \text{ nm}) = 1.96 \text{ eV}$$

In one sentence, explain why the excitation photon must have energy close to or greater than E_g to produce band-to-band PL. **(0.75 pts)**

The excitation photon must have energy close to or greater than Eg, so it has enough energy to traverse the bandgap, and then that allows it to later recombine via PL, causing the emission of a photon.

(b) Before analyzing the PL spectra, indicate whether emission is energetically possible based on the photon energy and bandgap. **(0.75 pts)**

Material	E_g **(eV)**	**405 nm (Y/N)**	**633 nm (Y/N)**
GaAs	1.424	Yes	Yes
InP	1.344	Yes	Yes
CdS	2.420	Yes	No
ZnO	3.200	No	No

(c) Identify the peak wavelength under each excitation condition. Enter "—" if no clear peak is observed. If there are two peaks, average their wavelength, then calculate

$$\boxed{E_{\text{PL}} = \frac{1240 \text{ nm} \cdot \text{eV}}{\lambda \text{ (nm)}}}$$

Compare with the bandgap energy E_g. **(3 pts)**

Material	λ_{405} **(nm)**	λ_{633} **(nm)**	λ_{avg} **(nm)**	E_{PL} **(eV)**	E_g **(eV)**	**% Error**
GaAs	870.6	871.5	871.0	1.424	1.424	0.03%
InP	922.2	922.5	922.3	1.344	1.344	0.03%
CdS	509.4	—	509.4	2.434	2.420	0.58%
ZnO	—	—	—	—	3.200	—

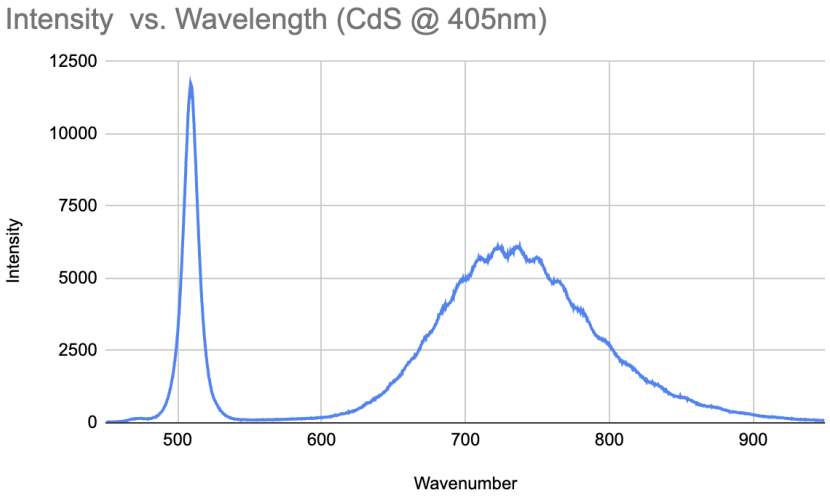


Figure 3: Sample Data Graphed for CdS with laser of 405 nm

i) GaAs and InP should show the same PL peak wavelength under 405 nm and 633 nm excitation. Why is emitted wavelength nearly independent of excitation wavelength? **For GaAs, the PL peaks are 870.6 nm under 405 nm excitation and 871.5 nm under 633 nm excitation. For InP, the peaks are 922.2 nm and 922.5 nm, respectively. The peak wavelengths are nearly the same because, after excitation, electrons quickly relax to the conduction-band minimum before recombining with holes. Therefore, the emitted photon energy is mainly determined by the semiconductor bandgap rather than by the excitation wavelength.**

ii) CdS shows PL under 405 nm excitation, but not 633 nm excitation. Why? **CdS shows a PL peak at 509.4 nm under 405 nm excitation, but no clear PL peak under 633 nm excitation. The 405 nm photon has an energy of about 3.06 eV, which is greater than the CdS bandgap of 2.42 eV, so it can excite electrons across the bandgap. In contrast, the 633 nm photon has an energy of about 1.96 eV, which is below the bandgap, so band-to-band excitation and PL do not occur.**

iii) Explain why ZnO shows no PL with either laser. Discuss photon energies and the bandgap of the material. **ZnO's gap is larger than both photon energies (3.06 eV and 1.96 eV), so neither laser can create free electron-hole pairs, and no band-edge PL (385–390 nm) is observed. Band-edge emission would also fall outside the measured range and behind the 405 nm edge filter. The broad 550 nm band seen under 405 nm excitation comes from excitation of defect states inside the gap, not from band-to-band recombination.**

Conclusion

GaAs and CdS were characterized with both absorption and photoluminescence. Use your results from Part A and Part B to complete the table.

Material	E_g **from absorption (eV)**	$E_{\mathbf{PL}}$ **from emission (eV)**	E_g **(eV)**
GaAs	1.393	1.424	1.424
CdS	2.377	2.434	2.420

(a) Why are the values of E_g from absorption and E_{PL} from emission close to one another? Explain how the measurements probe the same band structure with opposite optical processes. Give one reason the measured values may not be exactly identical, and one source of error during the experiment. Give your answer in 3–4 sentences. **(1 pt)**

Both methods probe the same band edges through opposite processes: in absorption, a photon with $h\nu \geq E_g$ promotes an electron from the valence band to the conduction band, while in PL the excited carriers thermalize to the band edges and recombine, emitting a photon of energy $\approx E_g$. The values are not exactly identical because PL comes from a thermal carrier population slightly above the band edge, whereas the absorption fit on thick samples only reaches $\alpha \sim 10^2\,\mathrm{cm}^{-1}$ and extrapolates from the sub-gap Urbach tail, lowering the estimate. One experimental source of error is the subjective choice of the linear fitting range in the absorption plot, along with assuming R is independent of wavelength.